Data Distributions

Data Distributions Dataset

- ‘titanic’ dataset in seaborn

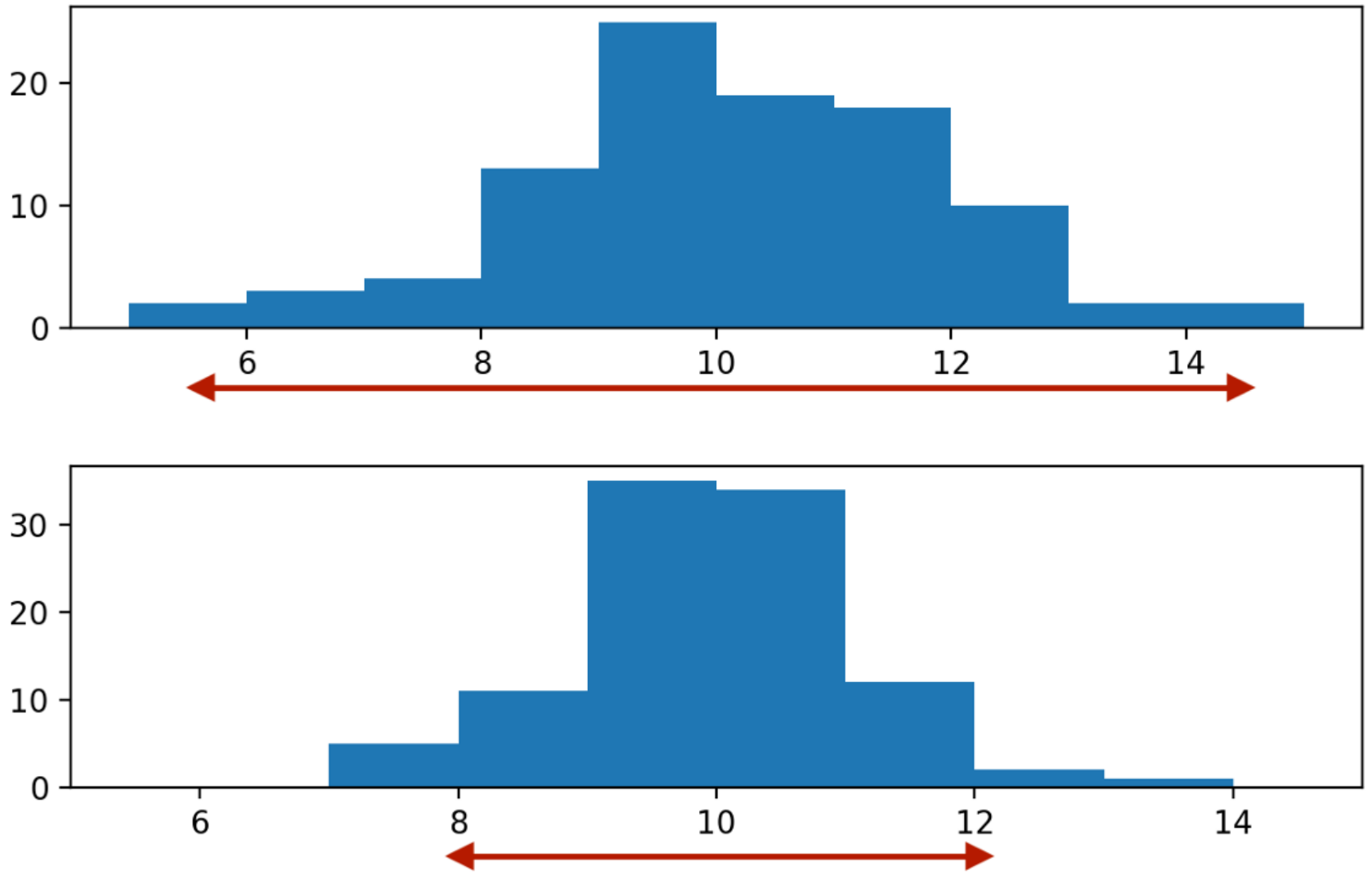
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
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Learning Objectives

By the end of this session, you will be able to:

- Understand the concept of data distributions
- Learn about skewness and kurtosis
- Apply Python code for calculating skewness and kurtosis
- Test your knowledge with quizzes and MCQs

What is Spread?



What is a Data Distribution?

A data distribution describes how values in a dataset are spread.

Types of distributions: Symmetric vs. Skewed

Examples: Normal, Uniform, Binomial, Poisson

Helps in: Statistical modeling, Machine learning, Data-driven decision-making

Types of Data Distributions

Distribution	Characteristics	Examples
Normal (Gaussian)	Bell-shaped, symmetric	Height, IQ scores
Uniform	All values equally likely	Dice rolls, Random numbers
Binomial	Success/failure trials	Coin flips, Pass/fail tests
Poisson	Rare events per time unit	Call center arrivals, Accidents
Exponential	Time until an event occurs	Service times, Waiting times

Real-World Applications of Distributions



FINANCE: STOCK RETURNS FOLLOW A NORMAL/LOG-NORMAL DISTRIBUTION



HEALTHCARE: DISEASE SPREAD FOLLOWS A POISSON DISTRIBUTION



RETAIL: CUSTOMER ARRIVALS FOLLOW AN EXPONENTIAL DISTRIBUTION



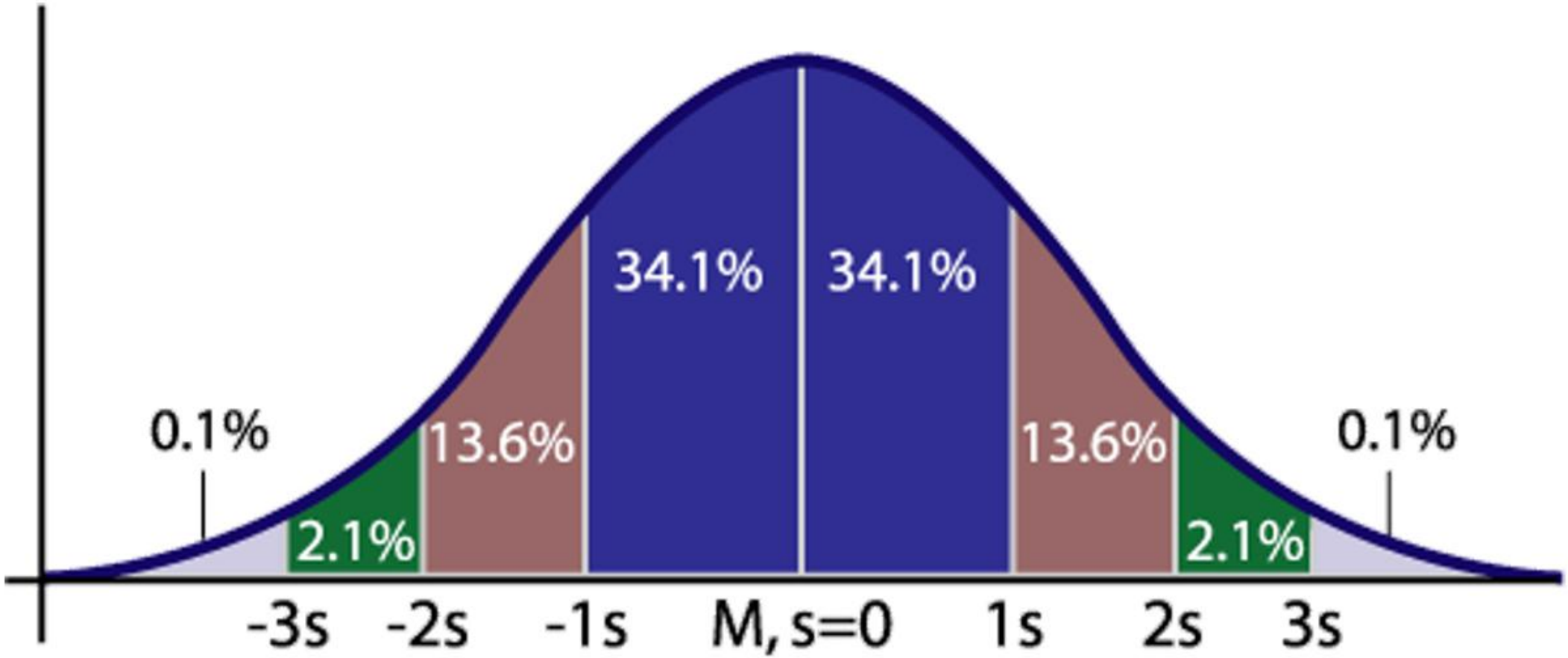
MANUFACTURING: QUALITY CONTROL USES A BINOMIAL DISTRIBUTION

Normal Distribution - The Bell Curve

- Symmetrical bell-shaped curve
- Defined by mean (μ) and standard deviation (σ)

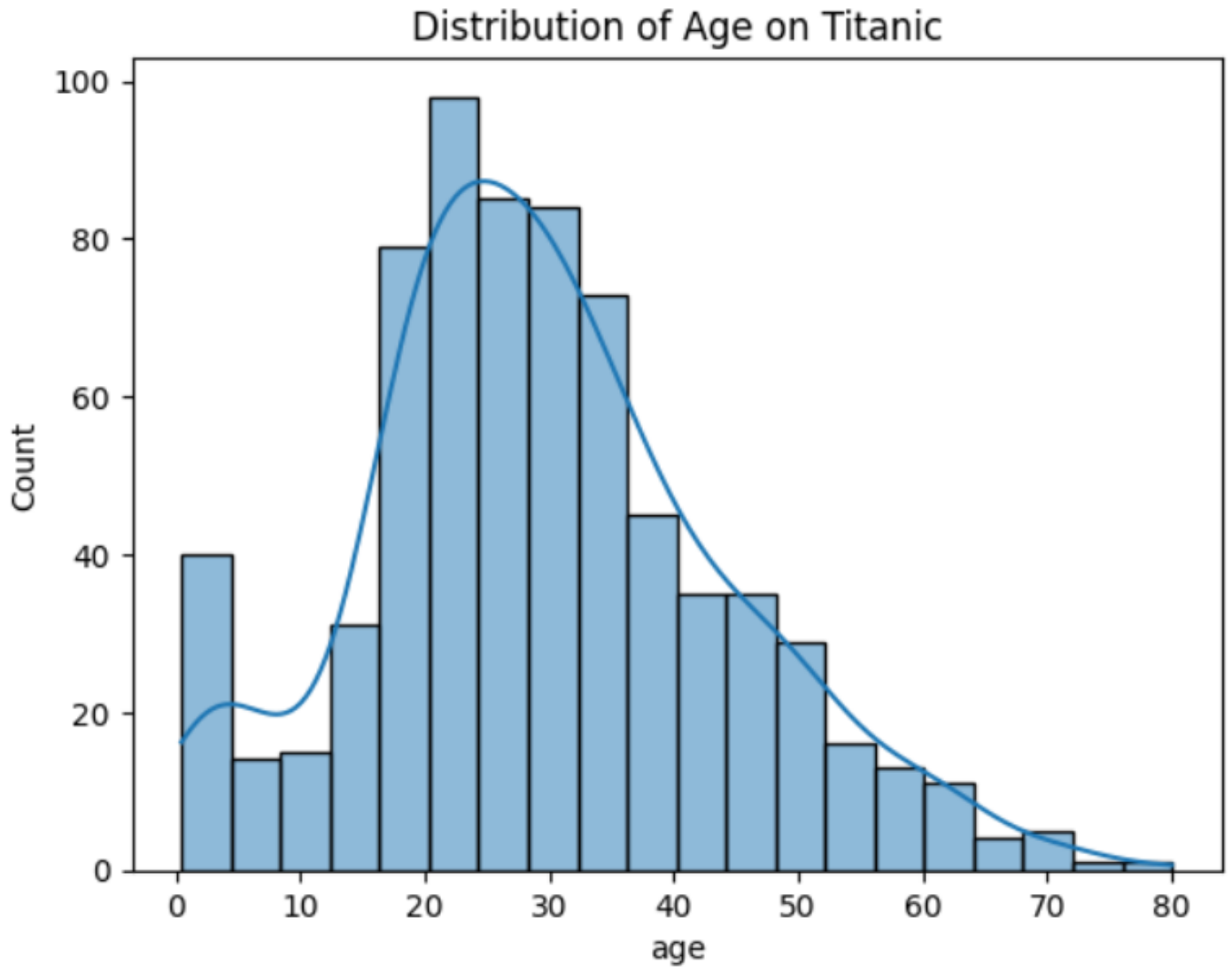
- 68-95-99.7 Rule:
    - 68% within 1σ
    - 95% within 2σ
    - 99.7% within 3σ

Normal Distribution - The Bell Curve



Visualizing Distribution

```
# Visualize the distribution of 'age'
sns.histplot(data['age'].dropna(), kde=True)
plt.title('Distribution of Age on Titanic')
plt.show()
```



Skewness - Symmetry of Data

Skewness tells us whether our data is skewed left, right, or symmetrical.

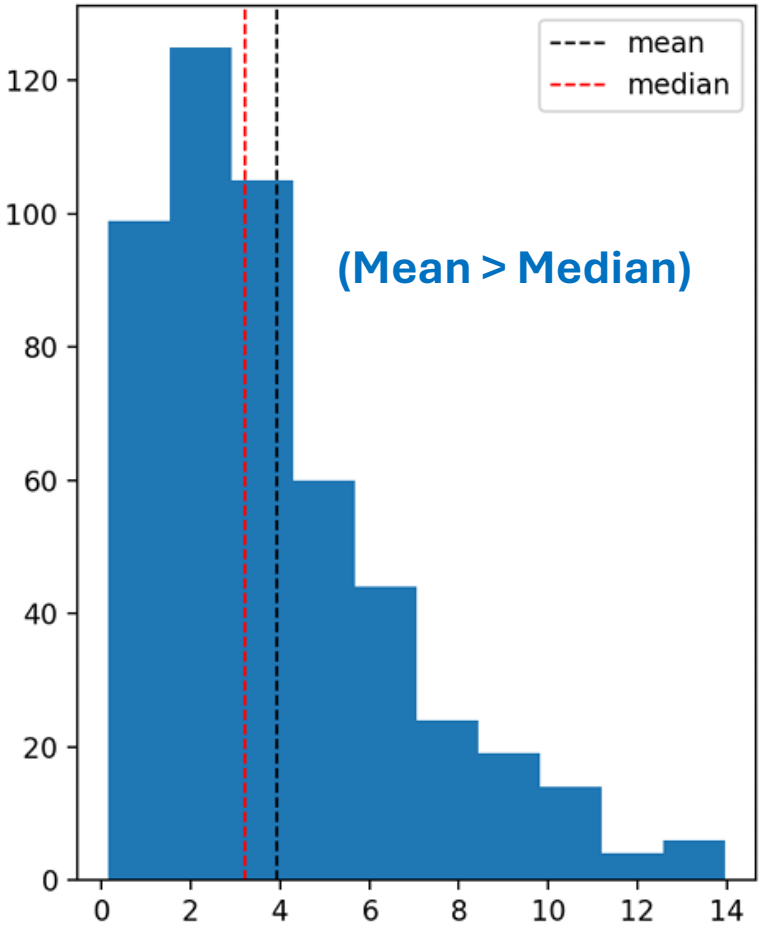
Measures asymmetry of distribution

- Formula:

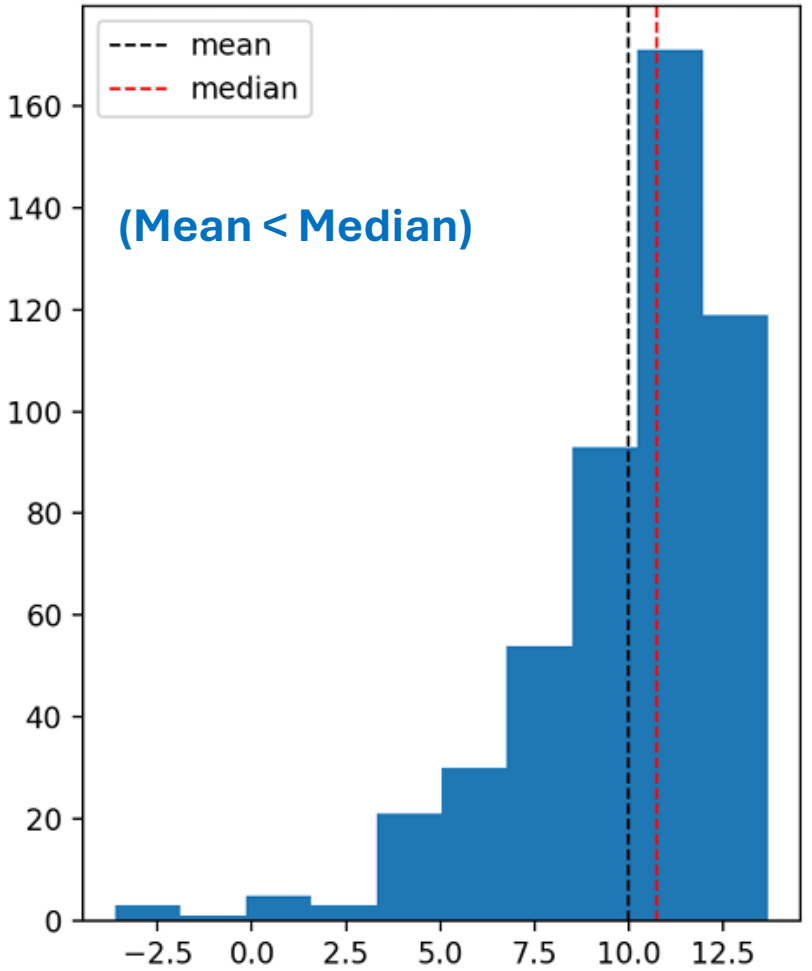
$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Skewness - Symmetry of Data

Right-skewed



Left-skewed



Skewness Example

- Given a dataset: [150, 200, 250, 300, 350, 400, 800], compute the **skewness**.

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Skewness Example

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$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Is the data right or left skewed?

Skewness Example

Dataset: [150, 200, 250, 300, 350, 400, 800]

- Mean:

$$\frac{150 + 200 + 250 + 300 + 350 + 400 + 800}{7} = 350$$

- **Median:** 300 (Middle value)
- **Skewness:** Since **Mean** > **Median**, data is **Right-Skewed**

Skewness in Python

```
# Calculate skewness
skewness_age = data['age'].skew()
skewness_fare = data['fare'].skew()

print(f"Skewness of Age: {skewness_age:.2f}")
print(f"Skewness of Fare: {skewness_fare:.2f}")
```

Explanation:

- **Positive skew (> 0):** Tail extends to the right.
- **Negative skew (< 0):** Tail extends to the left.
- **Zero skew:** Perfectly symmetrical.

Output:

```
Skewness of Age: 0.41
Skewness of Fare: 4.79
```

- **Age** is slightly positively skewed (some older passengers create a right tail).

- **Fare** is heavily positively skewed (some passengers paid extremely high fares).

Kurtosis - Tails of the Distribution

- Definition: **Kurtosis** measures whether data has **heavy or light tails**

- Types of kurtosis:

1. **Leptokurtic (High Kurtosis)**: Heavy tails (outliers). Kurtosis value > 3
2. **Mesokurtic (Normal Kurtosis)**: Like a normal distribution. Kurtosis value = 3 approx
3. **Platykurtic (Low Kurtosis)**: Light tails (less variation) Kurtosis value < 3

Formula:

$$\text{Kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

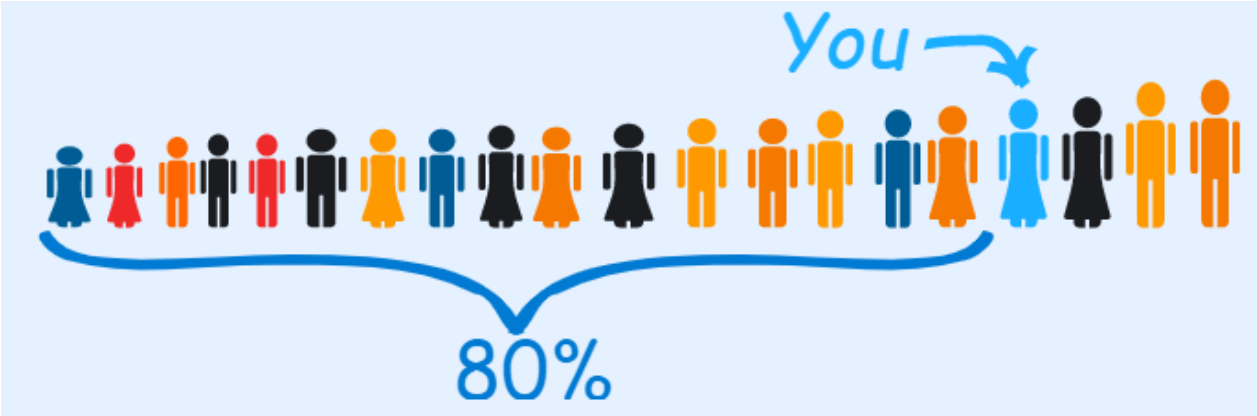
Skewness vs Kurtosis

- Skewness focusses on direction of outliers whereas kurtosis measures intensity of outliers.

- A dataset can be **symmetric (no skew)** but still have **high kurtosis** (many outliers)

Measures of Dispersion (IQR)

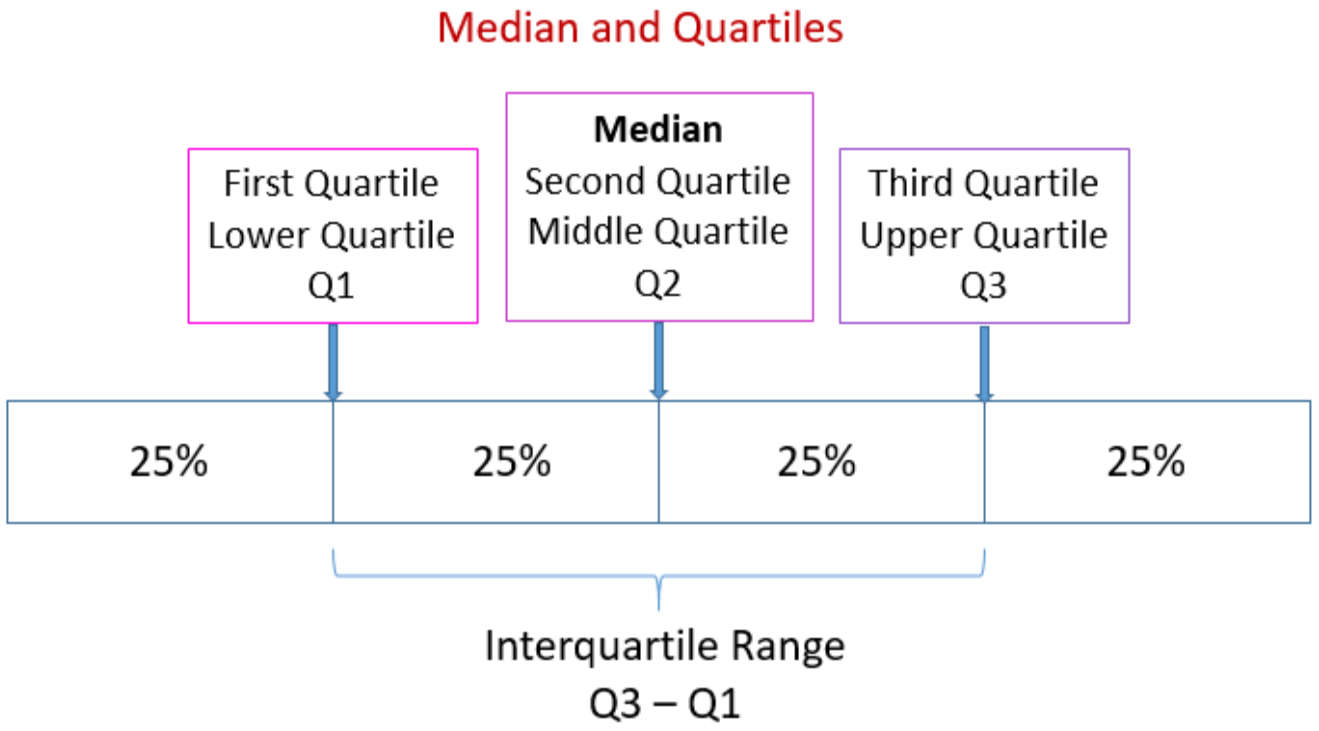
- **Percentile:** Percentiles indicate the percentage of scores that fall below a particular value.
- **Example:** Suppose you are the fourth tallest person in a group of 20 people. So that means 16 people are shorter than you. 80% of the people are shorter than you. That means you are at the 80th percentile.



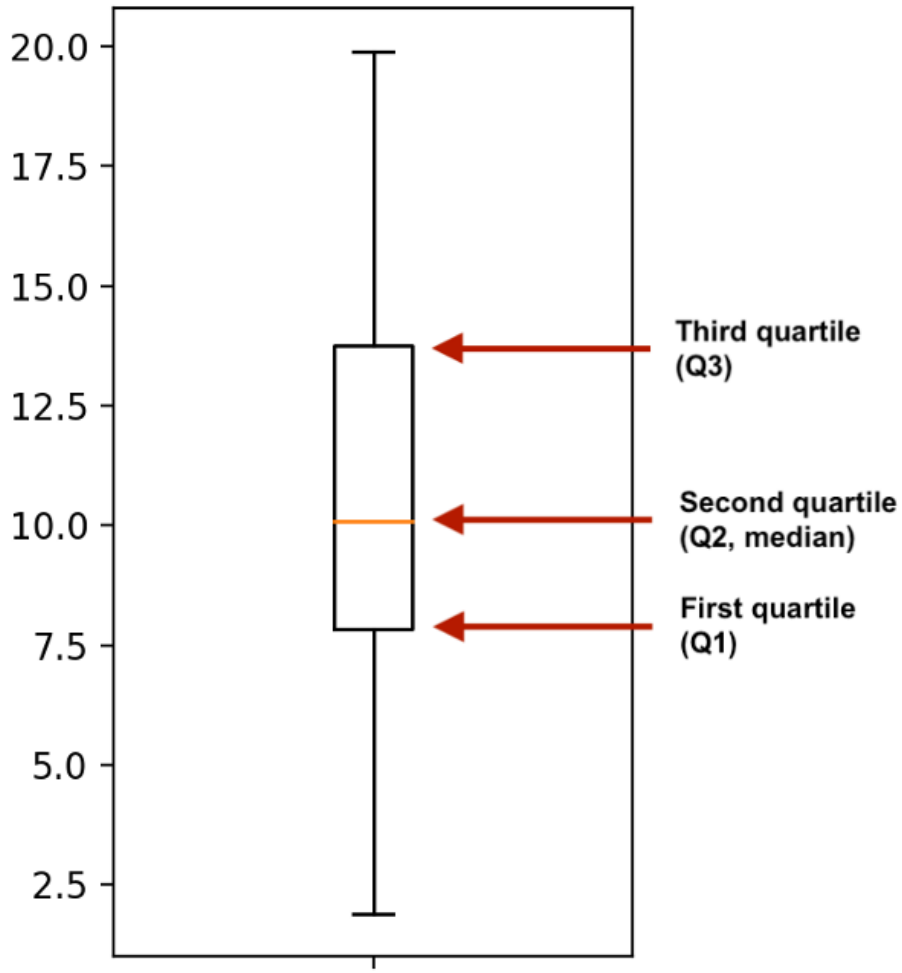
- Formula for Percentile = (Number of Values Below “x” / Total Number of Values) × 100

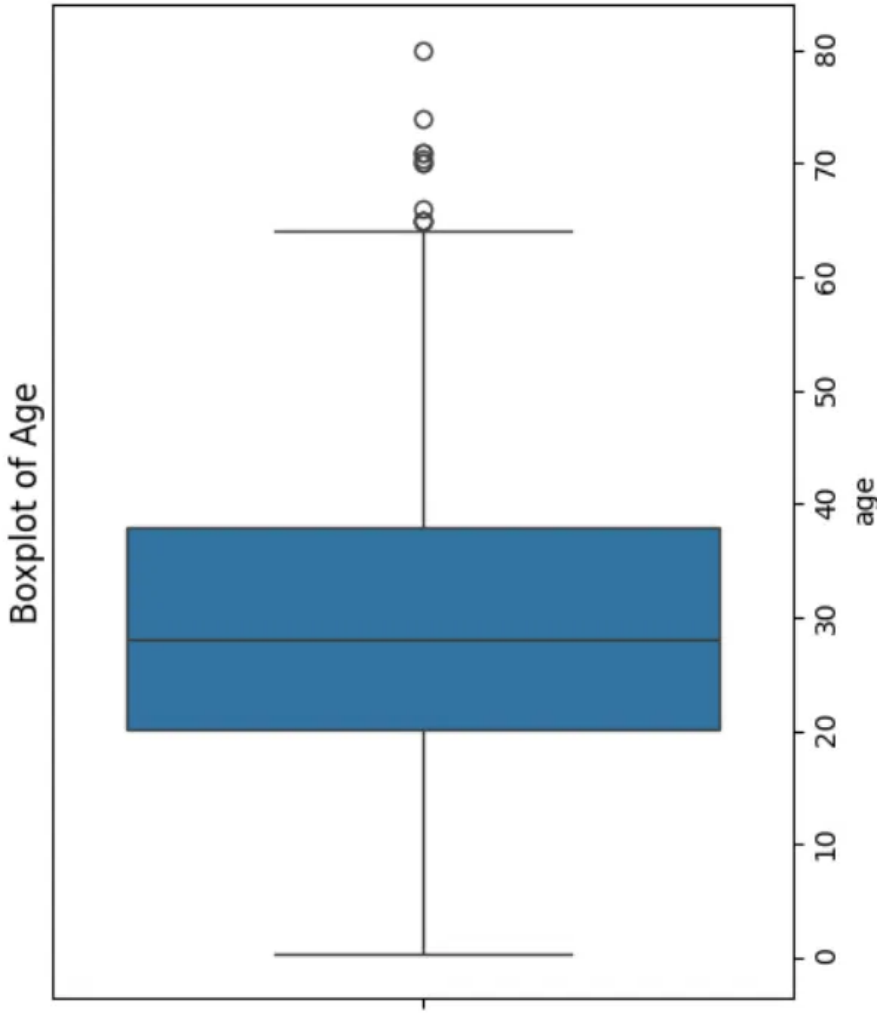
IQR (Interquartile range)

- **Quartiles:** Quartiles are the values that divide a list of numbers into quarters.
- Q1: Middle number between the smallest number and the median of the data set.
- Q2: Median of the data set.
- Q3: Middle value between the median and the highest value of the data set.
- Inter Quartile Range (IQR) = Q3 – Q1



Boxplot Representation





Python Implementation

https://github.com/vikashkmd/DSMLNotebooks/blob/main/Data%20distribution%20skewness%20and%20kurtosis.ipynb

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
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[The Ultimate Guide to Data Distributions: Skewness, Centering, and Spread Made Simple]

Multiple Choice Questions (MCQs)

Q1:What does a right-skewed distribution indicate?
A) The majority of values are low, with a few high values
B) The majority of values are high, with a few low values
C) The data is perfectly symmetrical
D) The data has no variation

Q2:Which measure is best for skewed data?
A) Mean
B) Median
C) Mode
D) Standard deviation

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A) Mean
B) Median ✅
C) Mode
D) Standard deviation

Q3 . What does a high positive kurtosis (leptokurtic distribution) indicate about a dataset?

A) The data has lighter tails and fewer extreme outliers than a normal distribution.

B) The distribution is skewed heavily to the right.

C) The data has heavy tails and a higher risk of extreme outliers.

D) The mean, median, and mode are all equal.

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Answer C)

In a negatively skewed (left-skewed) distribution, which of the following relationships generally holds true?

A) Mean > Median > Mode

B) Mean < Median < Mode

C) Mean = Median = Mode

D) Median > Mean > Mode

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C) Mean = Median = Mode
D) Median > Mean > Mode
Answer B)

Q. How does Winsorizing handle extreme outliers in a dataset?
A) It completely deletes the outlier observations from the dataset.
B) It replaces extreme values with the nearest non-outlier percentile values (e.g., 5th or 95th percentile).
C) It transforms the entire dataset using a logarithmic scale.
D) It replaces all missing and extreme values with the dataset mean.

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Q. How is the Interquartile Range (IQR) calculated, and why is it preferred over standard deviation for skewed data?
A) IQR = Q3 - Q1; it is robust to extreme outliers.
B) IQR = Q4 - Q1; it covers 100% of the dataset.
C) IQR = Q3 + Q1; it measures the total variance of the sample.
D) IQR = Mean - Median; it measures the symmetry of the data.

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Answer A)

Fill in the blanks

1. _____ measures the asymmetry of a data distribution.

2. The _____ rule states that 95% of data falls within 2 standard deviations of the mean in a normal distribution.

3. If the tails of a distribution are heavier than normal, it is called _____ kurtosis.

Fill in the blanks

1. _____ measures the asymmetry of a data distribution. (Answer: Skewness)

2. The _____ rule states that 95% of data falls within 2 standard deviations of the mean in a normal distribution. (Answer: 68-95-99.7 Rule)

3. If the tails of a distribution are heavier than normal, it is called _____ kurtosis. (Answer: Leptokurtic)

Session Summary

In this session, we have discussed about:

- Understand the concept of data distributions
- Learn about skewness and kurtosis
- Explore measures of central tendency
- Apply Python code for calculating skewness and kurtosis
- Test your knowledge with quizzes and MCQs

Reference Links

- Python notebook
- The Ultimate Guide to Data Distributions: Skewness, Centering, and Spread Made Simple
- How to Create Stunning Data Visualizations in Python: Top 10 Techniques to Learn
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- Phases of data science and analytics.